This document outlines how the team used GenerativeAI for help with the FiTech data driven card design model problem:

Following are the screenshots of some specific questions asked from chat gpt and how it respond to them-

Clv

		apr	fixed_var	annual_fee
1	0	14.9	Fixed	20
2	1	14.9	Fixed	20
3	2	14.9	Fixed	20
4	3	14.9	Variable	20
5	4	14.9	Variable	20
6	5	14.9	Variable	20

These are the results. Suggest how many emails to send

Your dataset includes the following relevant columns:

- **APR** (`apr`): The interest rate for the credit card.
- **Fixed/Variable Rate** (`fixed_var`): Whether the card has a fixed or variable interest rate.
- **Annual Fee** (`annual_fee`): Whether the card has a $20 fee or no fee.
- **BK Score** (`bk_score`): Customer risk category (150, 200, or 250).

GPT helped me analyze the dataset by summarizing key features like APR, fixed/variable rates, and annual fees.
It also provided insights on how to segment customers and determine an optimal email-sending strategy

We created an experiment test of 12 values out of 36 and created a logistic regression based on the experiment results. How do I decide how many emails to send for bk 150,200,250 to maximise profit

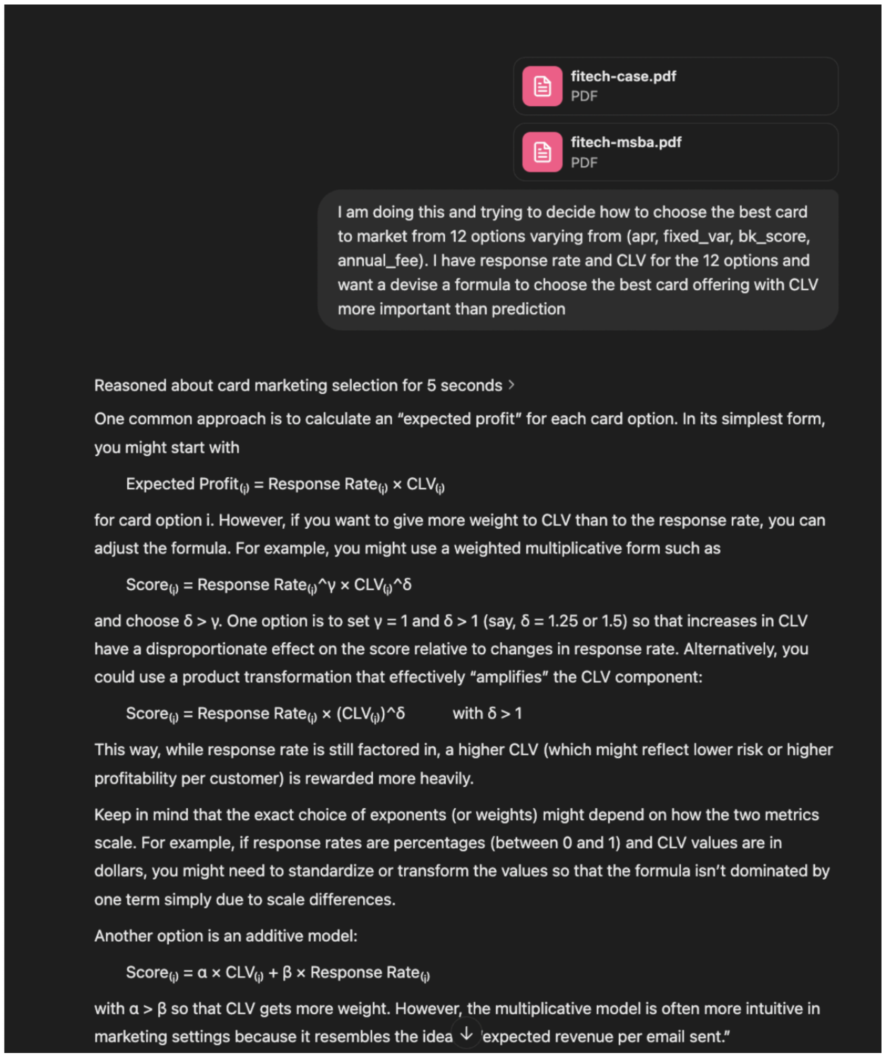
To maximize profit in your FiTech experiment after running the logistic regression model, follow these steps to determine the optimal number of emails to send for each BK score group (150, 200, 250):

Step 1: Understand the Model Output

Since you created a logistic regression model based on the experiment results, ensure you have:

- **Predicted response probability** for each product and BK score segment.

GPT helped interpret the logistic regression results by guiding on how to use predicted response probabilities and customer lifetime value (CLV) to determine the optimal number of emails to send.
It provided a structured approach to maximize profit by segmenting BK scores (150, 200, 250) and adjusting email strategies accordingly.



fitech-case.pdf
PDF

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PDF

I am doing this and trying to decide how to choose the best card to market from 12 options varying from (apr, fixed_var, bk_score, annual_fee). I have response rate and CLV for the 12 options and want a devise a formula to choose the best card offering with CLV more important than prediction

Reasoned about card marketing selection for 5 seconds ›

One common approach is to calculate an "expected profit" for each card option. In its simplest form, you might start with

$$\text{Expected Profit}_{(i)} = \text{Response Rate}_{(i)} \times \text{CLV}_{(i)}$$

for card option i. However, if you want to give more weight to CLV than to the response rate, you can adjust the formula. For example, you might use a weighted multiplicative form such as

$$\text{Score}_{(i)} = \text{Response Rate}_{(i)}\text{\^{}}\gamma \times \text{CLV}_{(i)}\text{\^{}}\delta$$

and choose $\delta > \gamma$. One option is to set $\gamma = 1$ and $\delta > 1$ (say, $\delta = 1.25$ or 1.5) so that increases in CLV have a disproportionate effect on the score relative to changes in response rate. Alternatively, you could use a product transformation that effectively "amplifies" the CLV component:

$$\text{Score}_{(i)} = \text{Response Rate}_{(i)} \times (\text{CLV}_{(i)})\text{\^{}}\delta \qquad \text{with } \delta > 1$$

This way, while response rate is still factored in, a higher CLV (which might reflect lower risk or higher profitability per customer) is rewarded more heavily.

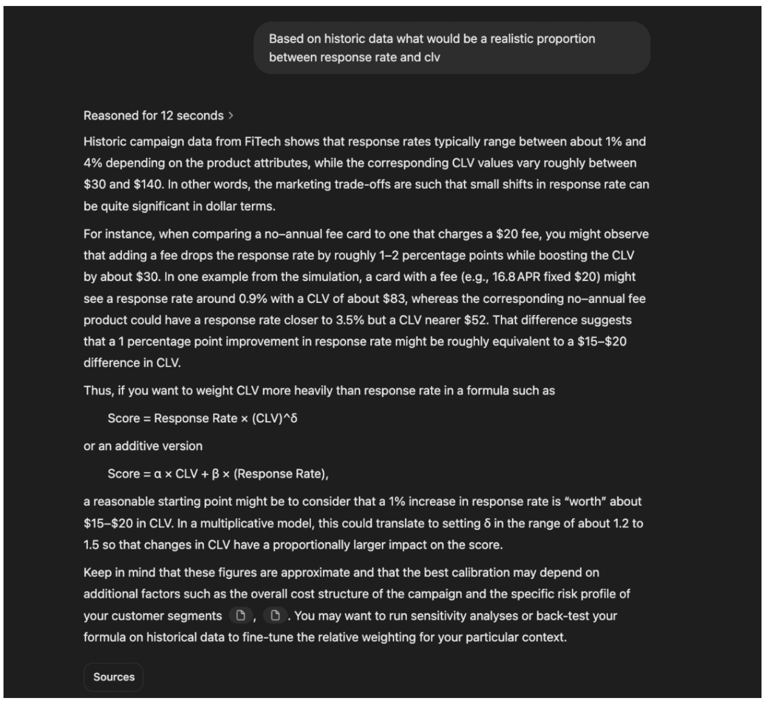
Keep in mind that the exact choice of exponents (or weights) might depend on how the two metrics scale. For example, if response rates are percentages (between 0 and 1) and CLV values are in dollars, you might need to standardize or transform the values so that the formula isn't dominated by one term simply due to scale differences.

Another option is an additive model:

$$\text{Score}_{(i)} = \alpha \times \text{CLV}_{(i)} + \beta \times \text{Response Rate}_{(i)}$$

with $\alpha > \beta$ so that CLV gets more weight. However, the multiplicative model is often more intuitive in marketing settings because it resembles the idea 'expected revenue per email sent."

GPT helped me come up with a solid approach for selecting the best credit card to market by suggesting a formula that balances response rate and CLV, with more weight on CLV since profitability per customer is key. It recommended both a multiplicative and an additive model, explaining how adjusting weights or exponents can impact the final decision. This made the selection process more data-driven and aligned with maximizing expected profit rather than just relying on predictions.



Based on historic data what would be a realistic proportion between response rate and clv

Reasoned for 12 seconds >

Historic campaign data from FiTech shows that response rates typically range between about 1% and 4% depending on the product attributes, while the corresponding CLV values vary roughly between $30 and $140. In other words, the marketing trade-offs are such that small shifts in response rate can be quite significant in dollar terms.

For instance, when comparing a no–annual fee card to one that charges a $20 fee, you might observe that adding a fee drops the response rate by roughly 1–2 percentage points while boosting the CLV by about $30. In one example from the simulation, a card with a fee (e.g., 16.8 APR fixed $20) might see a response rate around 0.9% with a CLV of about $83, whereas the corresponding no–annual fee product could have a response rate closer to 3.5% but a CLV nearer $52. That difference suggests that a 1 percentage point improvement in response rate might be roughly equivalent to a $15–$20 difference in CLV.

Thus, if you want to weight CLV more heavily than response rate in a formula such as

Score = Response Rate × (CLV)^δ

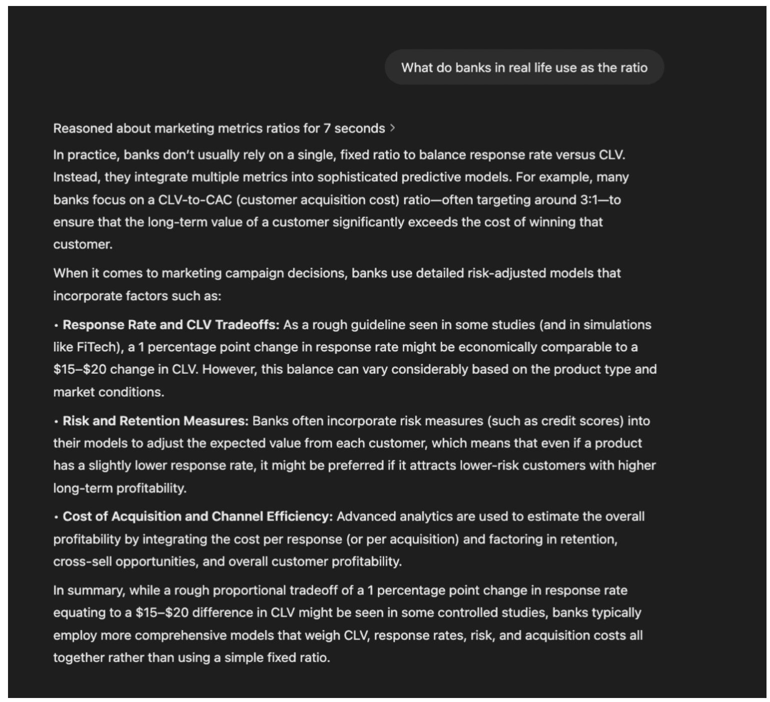
or an additive version

Score = α × CLV + β × (Response Rate),

a reasonable starting point might be to consider that a 1% increase in response rate is "worth" about $15–$20 in CLV. In a multiplicative model, this could translate to setting δ in the range of about 1.2 to 1.5 so that changes in CLV have a proportionally larger impact on the score.

Keep in mind that these figures are approximate and that the best calibration may depend on additional factors such as the overall cost structure of the campaign and the specific risk profile of your customer segments , . You may want to run sensitivity analyses or back-test your formula on historical data to fine-tune the relative weighting for your particular context.

Sources



What do banks in real life use as the ratio

Reasoned about marketing metrics ratios for 7 seconds >

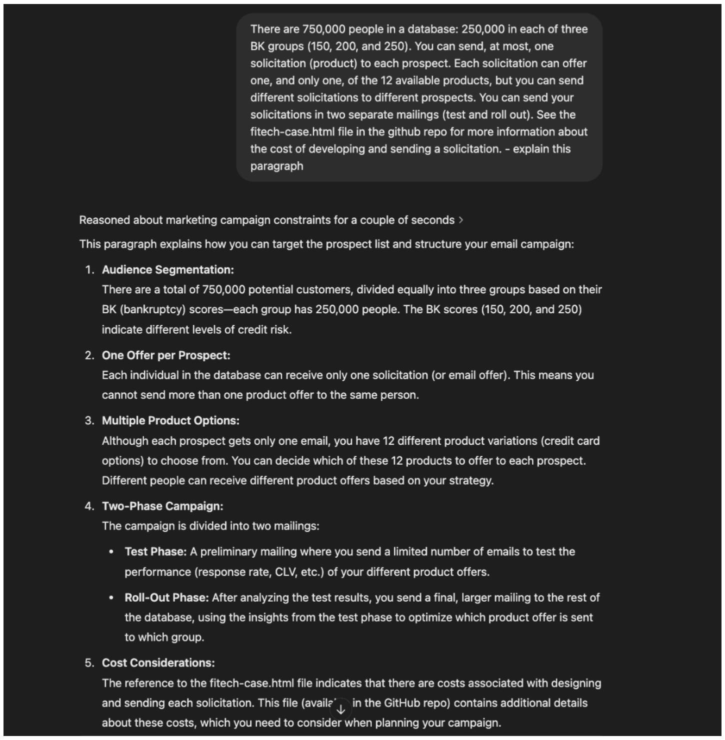
In practice, banks don't usually rely on a single, fixed ratio to balance response rate versus CLV. Instead, they integrate multiple metrics into sophisticated predictive models. For example, many banks focus on a CLV-to-CAC (customer acquisition cost) ratio—often targeting around 3:1—to ensure that the long-term value of a customer significantly exceeds the cost of winning that customer.

When it comes to marketing campaign decisions, banks use detailed risk-adjusted models that incorporate factors such as:

- **Response Rate and CLV Tradeoffs:** As a rough guideline seen in some studies (and in simulations like FiTech), a 1 percentage point change in response rate might be economically comparable to a $15–$20 change in CLV. However, this balance can vary considerably based on the product type and market conditions.
- **Risk and Retention Measures:** Banks often incorporate risk measures (such as credit scores) into their models to adjust the expected value from each customer, which means that even if a product has a slightly lower response rate, it might be preferred if it attracts lower-risk customers with higher long-term profitability.
- **Cost of Acquisition and Channel Efficiency:** Advanced analytics are used to estimate the overall profitability by integrating the cost per response (or per acquisition) and factoring in retention, cross-sell opportunities, and overall customer profitability.

In summary, while a rough proportional tradeoff of a 1 percentage point change in response rate equating to a $15–$20 difference in CLV might be seen in some controlled studies, banks typically employ more comprehensive models that weigh CLV, response rates, risk, and acquisition costs all together rather than using a simple fixed ratio.

GPT helped understand the relationship between response rate and CLV by explaining that, based on historic data, a 1% increase in response rate is roughly equivalent to a $15–$20 change in CLV. It also provided formulas to balance these factors, suggesting a weighted approach where CLV is prioritized. Additionally, it explained how banks use more sophisticated models, incorporating risk, acquisition costs, and retention to optimize marketing decisions, rather than relying on a fixed ratio. This helped me refine my decision-making framework for selecting the best card to market.

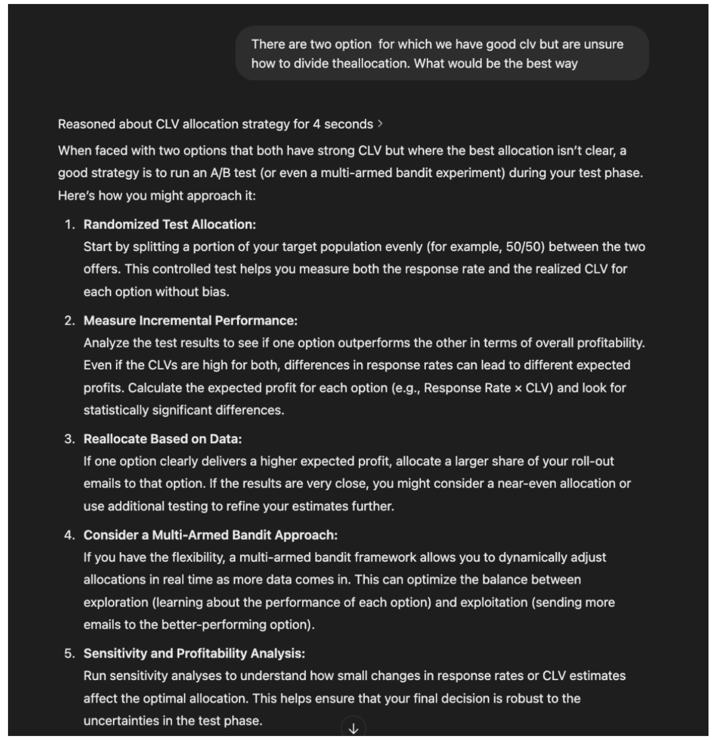


There are 750,000 people in a database: 250,000 in each of three BK groups (150, 200, and 250). You can send, at most, one solicitation (product) to each prospect. Each solicitation can offer one, and only one, of the 12 available products, but you can send different solicitations to different prospects. You can send your solicitations in two separate mailings (test and roll out). See the fitech-case.html file in the github repo for more information about the cost of developing and sending a solicitation. - explain this paragraph

Reasoned about marketing campaign constraints for a couple of seconds >

This paragraph explains how you can target the prospect list and structure your email campaign:

1. **Audience Segmentation:**
   There are a total of 750,000 potential customers, divided equally into three groups based on their BK (bankruptcy) scores—each group has 250,000 people. The BK scores (150, 200, and 250) indicate different levels of credit risk.
2. **One Offer per Prospect:**
   Each individual in the database can receive only one solicitation (or email offer). This means you cannot send more than one product offer to the same person.
3. **Multiple Product Options:**
   Although each prospect gets only one email, you have 12 different product variations (credit card options) to choose from. You can decide which of these 12 products to offer to each prospect. Different people can receive different product offers based on your strategy.
4. **Two-Phase Campaign:**
   The campaign is divided into two mailings:
   - **Test Phase:** A preliminary mailing where you send a limited number of emails to test the performance (response rate, CLV, etc.) of your different product offers.
   - **Roll-Out Phase:** After analyzing the test results, you send a final, larger mailing to the rest of the database, using the insights from the test phase to optimize which product offer is sent to which group.
5. **Cost Considerations:**
   The reference to the fitech-case.html file indicates that there are costs associated with designing and sending each solicitation. This file (avail in the GitHub repo) contains additional details about these costs, which you need to consider when planning your campaign.



There are two option for which we have good clv but are unsure how to divide theallocation. What would be the best way

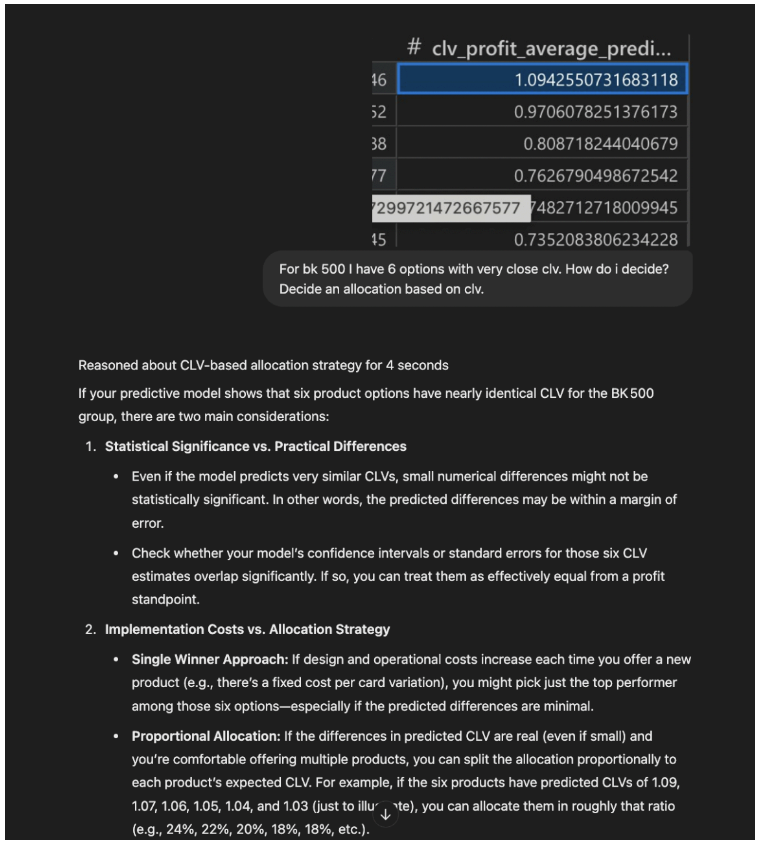
Reasoned about CLV allocation strategy for 4 seconds >

When faced with two options that both have strong CLV but where the best allocation isn't clear, a good strategy is to run an A/B test (or even a multi-armed bandit experiment) during your test phase. Here's how you might approach it:

1. **Randomized Test Allocation:**
   Start by splitting a portion of your target population evenly (for example, 50/50) between the two offers. This controlled test helps you measure both the response rate and the realized CLV for each option without bias.
2. **Measure Incremental Performance:**
   Analyze the test results to see if one option outperforms the other in terms of overall profitability. Even if the CLVs are high for both, differences in response rates can lead to different expected profits. Calculate the expected profit for each option (e.g., Response Rate × CLV) and look for statistically significant differences.
3. **Reallocate Based on Data:**
   If one option clearly delivers a higher expected profit, allocate a larger share of your roll-out emails to that option. If the results are very close, you might consider a near-even allocation or use additional testing to refine your estimates further.
4. **Consider a Multi-Armed Bandit Approach:**
   If you have the flexibility, a multi-armed bandit framework allows you to dynamically adjust allocations in real time as more data comes in. This can optimize the balance between exploration (learning about the performance of each option) and exploitation (sending more emails to the better-performing option).
5. **Sensitivity and Profitability Analysis:**
   Run sensitivity analyses to understand how small changes in response rates or CLV estimates affect the optimal allocation. This helps ensure that your final decision is robust to the uncertainties in the test phase.

GPT helped us structure our marketing campaign and optimize email allocation by:

1. Understanding Campaign Constraints: It clarified audience segmentation (750,000 prospects in three BK score groups), email limitations (one offer per person), and the two-phase approach (test phase followed by a rollout).
2. Cost Considerations: It highlighted cost factors from the GitHub file, helping us assess the financial impact of email solicitations.
3. Optimizing Allocation: It suggested using an A/B test or a multi-armed bandit approach to divide emails between two strong CLV options, analyzing response rates, and reallocating based on performance.
4. Ensuring Robust Decisions: It recommended sensitivity analysis to account for variations in response rates and CLV, ensuring a data-driven allocation strategy.



GPT helped us decide how to allocate emails among six product options with similar CLV by:

1. Assessing Statistical vs. Practical Differences: It advised checking confidence intervals and margin of error to determine if CLV differences are significant or just noise.
2. Balancing Implementation Costs: If operational costs increase per product variation, a single winner approach (choosing the top performer) may be best.
3. Using Proportional Allocation: If differences in CLV are meaningful, it suggested distributing emails in proportion to predicted CLVs (e.g., 24%, 22%, 20%, etc.).

This approach helped us make a data-driven and cost-effective allocation decision.

How it helped / hindered our understanding:

GPT greatly assisted our understanding by breaking down complex marketing and data-driven decision-making concepts, providing structured approaches for campaign design, and suggesting practical formulas for expected profit calculations. It also explained real-world applications, helping us grasp the relevance of our assignment. However, its limitations included a lack of context-specific insights, occasional over-simplification of concepts, and an inability to perform direct data analysis, requiring us to validate and adapt its recommendations. Overall, GPT was a valuable tool for structuring our approach, but critical evaluation was necessary to tailor its insights to our specific dataset and business context.

How our learning process was complemented:

GPT complemented our learning by guiding our decision-making, bridging theory with real-world applications, and accelerating problem-solving. It provided structured insights on key marketing strategies like A/B testing, CLV-based allocation, and multi-armed bandit approaches, helping us apply these concepts effectively. While it streamlined our research and discussions, it also encouraged critical thinking, as we had to validate its suggestions against our dataset. Overall, GPT acted as a valuable reference tool, enhancing collaboration and helping us develop a structured, data-driven approach to our assignment.